A quartic in 389 variables from local nonvanishing

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Abstract

We specify an integer quartic in 389 total variables, with 6198 nonzero monomials, by 361 quadratic equations and a complete coefficient table. Its rational input projection is the integers under the same arithmetic premises as the specified 392-variable construction. Three rational inverse witnesses can be omitted: at every prime where a local divisibility conclusion is required, the remaining equations already force their denominators to be nonzero. The additional fibers have empty contact radicals, for which the required divisibility statement is vacuous. The argument explicitly covers the degenerate value at which the detector's rational formula has a zero denominator. The polynomial has content one and coefficient height less than $2^{1791357}$. The largest coefficient is exactly three smaller than that of the starting polynomial.

1 The specified polynomial and its arithmetic scope

For $F \in \mathbb{Z}[X, Y_1, \dots, Y_m]$, its rational input projection is

$$\{x \in \mathbb{Q} : \exists \boldsymbol{y} \in \mathbb{Q}^m,\ F(x, \boldsymbol{y}) = 0\}.$$

Every variable count includes X. Degree means total degree, and coefficient height means the largest absolute integer coefficient after the fixed constants have been substituted.

Theorem 1.1. *The files* `quadratic-system389.json` *and* `polynomial389.json` *specify 361 integer quadratic polynomials Q_i in 389 coordinates and their sum of squares $P_{389} = \sum_i Q_i^2$. Its degree is exactly four, its integer content is one, and it has 6198 nonzero coordinate monomials. Its largest coefficient has 1791357 binary digits.*

Under the arithmetic premises of the 392-variable construction [1], its rational input projection is exactly $\mathbb{Z}$. No additional prime-pattern or arithmetic existence assertion is used.

The retained arithmetic premises are the specified elliptic-group and index-height assertions, the contact-height theorem, the pole obstruction and simultaneous pole realization, and Sun's simultaneous prime-pattern theorem with the quantitative weighted selection premise used in the quartic construction. These hypotheses are stated precisely in [1, Section 2]. The present theorem proves a new implication from those premises. The coefficient checks do not establish the deep arithmetic inputs.

The constant in the coefficient table is

$$A = 80989^{54929}.$$

It is an integer coefficient, not an additional coordinate. The starting polynomial is fixed by commit

`de7aa60dd1794d7ea62a8a83cdc4bfd22380c74f`

of the repository in [1]. Its full quadratic system and coefficient table are included as exact compressed source files.

2 The detector equations

Put $K = \mathbb{Q}(\rho)$, where $\rho^2 = 2$, and let σ denote its nontrivial automorphism. For $W = w_0 + w_1\rho$ one has

$$\mathrm{N}(W) = w_0^2 - 2w_1^2.$$

In particular, $\mathrm{N}(W) = 0$ if and only if $W = 0$.

For the three retained height detectors write

$$a \in \{1, C_0^2, (1 + C_1^2)^2\}, \qquad t = a\lambda, \qquad \delta = t - 1,$$

where $C_0 + C_1\rho$ is the product of the four root abscissae. The retained root charts and their real embeddings give $C_0 > 0$; all three values of a are therefore nonzero rational squares. The stored coordinates use affine substitutions. To remove any ambiguity, if u and v are the coordinates named `heightProduct.Z.0` and `heightProduct.Z.1`, then

$$C_0 = 2u, \qquad C_1 = 4v.$$

The stored quantities named `scale0`, `x1Square`, and `scale1` are u^2, v^2, v^4. Their consumers, rather than these names alone, recover the actual scales. Specifically, the stored coordinates t_1^*, t_2^* satisfy

$$t_1^* = \lambda u^2, \qquad t_2^* = \lambda v^2 + 8\lambda v^4,$$

and the actual inputs are

$$t_0 = \lambda, \qquad t_1 = 4t_1^* = C_0^2\lambda, \qquad t_2 = \lambda + 32t_2^* = (1 + C_1^2)^2\lambda.$$

These identities follow by direct substitution into the retained quadratic rows. The independent normalization checker verifies them as polynomial identities, together with the weight and pole-input identities below. Set

$$\Theta = \left\{0, -\tfrac{1}{2}, -1, \tfrac{1}{2399}, -\tfrac{32}{407}\right\}, \qquad F(t) = \prod_{\theta \in \Theta}(t - \theta).$$

Let $M(t)$ be the product of the primes $p > 76176$ for which $v_p(t - \theta)$ is positive and odd for at least one $\theta \in \Theta$.

With these actual coordinates, the retained quadratic gates imply the polynomial identity

$$F(t)e = (a\delta)^5. \tag{1}$$

Whenever $\delta F(t) \neq 0$, put

$$R = \frac{F(t)}{a^5\delta^6}, \qquad q_m = \frac{R^{1-2m}}{\delta^{2m+2}}, \qquad W = \prod_j (1 + q_{m_j} D_j^2) = w_0 + w_1\rho.$$

Here the $D_j \in K$ are the retained field-valued point, comparison, input, and Lucas tests. Their assigned orders are $m_j \in \{1, 10, 40\}$. The grouped products and power gates of the starting system compute exactly this product. Their definitions are polynomial identities; no deleted inverse witness is used by these gates.

The final detector rows, apart from its pole certificate, are

$$w_0^2 - 2w_0T - 2w_1^2 = 0, \qquad e\delta b = T^2. \tag{2}$$

The starting system additionally imposed $w_0 I = 1$ with a private rational inverse witness I. We delete this last row and this coordinate in each detector. The five normalized pole rows are retained. Whenever $b \neq 0$, they imply the same obstruction as the original pole equations: $v_p(b)$ cannot be negative and odd for $p \notin \{2, 3, 17\}$.

3 Local nonvanishing without inverse witnesses

Lemma 3.1. *For every rational solution of the modified system, $F(t) \neq 0$. If $p \mid M(t)$, then $\delta \neq 0$, $v_p(\delta) = 0$, and $v_p(R)$ is odd. If $\delta = 0$, then $M(t) = 1$.*

Proof. If $F(t) = 0$, then $t \in \Theta$, and hence $\delta \neq 0$ because $1 \notin \Theta$. This contradicts (1), since $a \neq 0$.

For $p > 76176$, every nonzero difference of two elements of $\Theta \cup \{1\}$ is a p-adic unit. This follows by direct factorization of their rational numerators and denominators. Thus a prime in $M(t)$ divides exactly one factor of $F(t)$ to positive odd order, while δ is a unit. Since a is a rational square,

$$v_p(R) = v_p(F(t)) - 5v_p(a)$$

is odd. Its sign has not yet been assumed.

Finally $\delta = 0$ gives $t = 1$. The five differences $1 - \theta$ have no numerator or denominator prime greater than 76176. Consequently $M(1) = 1$. □

Lemma 3.2. *If $M(t) > 1$, then W, w_0, T, b are nonzero and the retained detector equations have their intended rational interpretation. In particular,*

$$T = \frac{1}{\mathrm{Tr}(W^{-1})}, \qquad b = \frac{R}{\mathrm{Tr}(W^{-1})^2}.$$

Proof. Choose $p \mid M(t)$ and put $\nu = v_p(R)$, an odd integer by Lemma 3.1. The prime p is unramified in K. At every place v above p, normalized by $v(p) = 1$, its value group on $K^\times$ is $\mathbb{Z}$. Since δ is a unit,

$$v(q_m D^2) = (1 - 2m)\nu + 2v(D)$$

is odd whenever $D \neq 0$. It cannot be the valuation of -1. Therefore every factor $1 + q_m D^2$ is nonzero, also when $D = 0$, and hence $W \neq 0$.

If $w_0 = 0$, the first equation of (2) gives $w_1 = 0$, contrary to $W \neq 0$. Thus $w_0 \neq 0$. Its norm is nonzero, and the same equation gives

$$T = \frac{\mathrm{N}(W)}{2w_0} = \frac{1}{\mathrm{Tr}(W^{-1})} \neq 0.$$

Moreover (1) gives $e\delta = 1/R \neq 0$. The second equation of (2) now yields $b = RT^2 \neq 0$. All divisions are justified by the remaining equations and the chosen contact prime; none uses the deleted inverse row. □

Proposition 3.3. *For every $p \mid M(t)$ there is a place v above p at which*

$$v(D_j) \geq m_j \quad \textit{for every } j.$$

The asserted divisibility consequences of the starting detector remain valid when $M(t) = 1$.

Proof. By Lemma 3.2 the pole input and the norm-trace formula are defined and nonzero. For each $v \mid p$, parity gives

$$v(1 + q_{m_j} D_j^2) = \min\{0, \nu + 2(v(D_j) - m_j\nu)\} \leq 0.$$

Put $\ell = \max_{v|p} v(W) \leq 0$. The trace inequality implies

$$v_p(\mathrm{Tr}(W^{-1})) \geq -\ell, \qquad v_p(T) \leq \ell.$$

The valuation of $b = RT^2$ is odd. The pole obstruction therefore implies

$$0 < v_p(b) \leq \nu + 2\ell.$$

In particular $\nu > 0$ and $\ell > -\nu/2$. At a place attaining ℓ, every factor valuation is at least ℓ, because all the factor valuations are nonpositive and their sum is ℓ. Consequently

$$v(D_j) > (m_j - 3/4)\nu \geq m_j - 3/4.$$

The value group is integral, so $v(D_j) \geq m_j$.

If $M(t) = 1$, every conclusion of the form $M(t)^k \mid N$ is automatic. No application of a rational pole formula is needed for that detector in this case. This includes $t = 1$, for which $h(t) = 0$. □

Remark 3.4. The argument is local in its use of nonvanishing but proves a global integer-projection statement. It does not assert that every new rational witness tuple lifts to one of the old tuples: a detector with empty contact radical can have $W = 0$. Such a tuple is handled by the vacuous divisibility conclusion, not by an undefined reciprocal.

4 Soundness and completeness

Proof of the projection assertion in Theorem 1.1. All retained point charts, scalar equations, the forced-prime normalizer, and the scalar nonzero guard are unchanged. The three deleted coordinates occur only in their respective inverse rows. The arithmetic point interpretations and the fixed-prime index estimate therefore remain valid before any use of a height detector.

Proposition 3.3 gives exactly the local test orders required at every prime in each of the three contact radicals. The formal logarithm comparison, integer-label congruences and the two height contradictions in [1, Sections 2 and 5] use these conclusions. They do not require a detector denominator at primes outside its contact radical. They are consequently unchanged. The contact-height bound itself is a statement about the rational number t; it applies also when its contact radical is one. By Lemma 3.1, none of its excluded contact values occurs.

In particular the unchanged argument proves that the Lucas residual is zero and then that the input label

$$\beta = 51(2X + 1)^2 + 2504$$

equals its odd integer elliptic index. If $2X + 1 = u/v$ is in lowest terms with $v > 0$, this forces $v^2 \mid 51$, so $v = 1$; oddness of β then makes u odd. Hence $X \in \mathbb{Z}$.

Conversely every integer input has a solution of the 392-variable system under the same arithmetic premises. Forgetting the three private inverse coordinates gives a solution of the new system. There are no new constraints and no new simultaneous-choice problem. Finally a sum of rational squares is zero exactly when every square is zero, proving the assertion for P_{389}. □

5 Exact data and independent verification

The source coordinate numbers and residual numbers are inherited from the upstream arithmetic circuit. Their deletion is the complete change:

Source coordinate	Coordinate in the 392 system	Source row
382	221	334
430	250	384
478	279	434

Each deleted row is $1 - w_0 I = 0$. Every other primitive integer residual is retained verbatim, apart from renumbering its coordinates in increasing order. Thus the counts are $392 - 3 = 389$ coordinates and $364 - 3 = 361$ quadratic residuals.

The producer `build_quartic389.py` performs this deletion and expands the complete sum of squares. The separate program `check_quartic389.py` imports no producer. It reconstructs the original 392-variable polynomial, checks the sole-use inverse coordinates and every retained row, and compares the entire new coefficient table in two ways: direct squaring and subtraction of the three deleted guard squares.

The additional program `verify_detector_normalizations.py` independently substitutes the complete affine coordinate dictionary into the relevant residuals, using exact rational arithmetic. The three weight rows become $F(t)e - (a\delta)^5$ multiplied, respectively, by 1952786, 976393/8, and 1952786. The retained rows also give $\delta^3 q_1 = e$ and $e\delta b = T^2$. The toric power equations give $q_{10} = e^{19}/\delta^3$ and $q_{40} = e^{79}/\delta^3$ on their nonzero domain. Thus the affine substitutions do not change the semantic scales, weights, or pole inputs.

There are 6219 terms when powers of the fixed integer A are listed separately, and 6198 ordinary nonzero coordinate monomials after equal coordinate monomials are collected. Exactly six nonconstant monomials disappear. The constant coefficient decreases by three. Its unchanged leading term is

$$11082847123849826043664 \cdot A^2.$$

Exact integer comparison proves that the constant coefficient is still the largest coefficient in absolute value and still has 1791357 binary digits. A retained coefficient equal to one gives content one; a retained nonzero quartic monomial gives degree exactly four.

Run the following commands in the construction directory:

```
python3 build_quartic389.py
python3 check_quartic389.py
python3 verify_detector_normalizations.py
```

The accompanying hashes bind the complete polynomial and quadratic system. The finite computation checks the algebra and the asserted numerical invariants. The local argument and its arithmetic inputs are stated separately above.

References

[1] J. Brox and contributors, *A forced prime condition and an explicit quartic in 392 variables*, 8 November 2026, source and computational data at commit `de7aa60dd1794d7ea62a8a83cdc4bfd22380c74f`, https://github.com/JoseBrox/diophantine-z-q_improvements.

[2] E. Glazer and contributors, *An explicit quartic defining the integers in the rationals*, Version 13, commit `5bbd74f6a479776a332cbe036809d5dddae63720`, https://github.com/elliotglazer/diophantine-z-q.